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ON

Comparative Analysis of Spatial, Edge-Preserving and Transform-Based Filters for Medical Image Enhancement

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In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled "Comparative Analysis of Spatial, Edge-Preserving and Transform-Based Filters for Medical Image Enhancement" in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of our faculty guide. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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ABSTRACT

Medical imaging modalities such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT) and ultrasound are widely used for clinical diagnosis and treatment planning. However, images obtained from these systems are often degraded by noise introduced during acquisition, transmission, or reconstruction processes. Effective denoising is therefore essential to enhance image quality while preserving important anatomical structures and diagnostic details.

This study presents a comprehensive comparative analysis of multiple medical image denoising techniques, including classical spatial filters, edge-preserving methods and transform-domain approaches. The evaluated techniques include Mean, Median, Gaussian, Wiener, Bilateral and Anisotropic Diffusion filters, as well as advanced approaches such as Non-Local Means, Wavelet Transform, Curvelet Transform, Moving Average, Exponential Moving Average and a hybrid Median-Mean filtering strategy. The performance of these techniques was quantitatively assessed using widely adopted image quality metrics, including Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM) and Signal-to-Noise Ratio (SNR).

Experimental results demonstrate that among spatial filters, the Median filter performs best for MRI images while the Mean filter excels for ultrasound. The Bilateral filter consistently outperforms other edge-preserving methods across all modalities. Among transform-based approaches, the Wiener filter achieves the most reliable performance. Anisotropic diffusion exhibits poor results across all datasets. The findings of this study offer valuable insights for selecting suitable denoising techniques for different medical imaging modalities and clinical applications.

Keywords: Medical Image Denoising, Median Filter, Gaussian Filter, Anisotropic Diffusion, Wavelet Transform, Curvelet Transform, PSNR, SSIM, Medical Imaging, Bilateral Filter, Non-Local Means, Wiener Filter

1. Introduction

Medical imaging plays a crucial role in disease diagnosis, surgical planning, and treatment monitoring. Modalities such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT) and ultrasound generate high-resolution anatomical images that assist radiologists and clinicians in identifying pathological conditions with precision. However, medical images are often contaminated by various types of noise such as Gaussian noise, salt-and-pepper noise and speckle noise. Noisy images pose a significant challenge for healthcare professionals, as noise can obscure important diagnostic information and potentially affect the accurate identification of diseases such as tumors.

Noise in medical images can be defined as an unwanted and random signal that degrades image quality. It may arise from multiple sources, including electronic fluctuations in imaging sensors, transmission errors, or the inherent physical processes underlying each imaging modality. In MRI systems, noise typically originates from thermal fluctuations in receiver electronics and manifests as Rician or Gaussian distributed noise. CT images primarily contain Gaussian noise from photon detection and signal processing. Ultrasound images are predominantly affected by speckle noise, which results from the constructive and destructive interference of reflected acoustic waves within biological tissues.

Over the years, numerous denoising techniques have been proposed, ranging from traditional linear smoothing filters to advanced transform-domain and non-local approaches. These methods aim to improve image quality while minimizing the loss of critical structural and diagnostic information. However, selecting an appropriate denoising technique remains a challenging task because different methods exhibit varying performance depending on the type of noise, imaging modality and computational constraints. For example, spatial filters are computationally simple but may cause edge blurring, whereas transform-domain and non-local approaches provide better structural preservation at the cost of higher computational complexity.

Therefore, a systematic comparative evaluation of different denoising techniques is necessary to determine their effectiveness for specific medical imaging modalities. This project report presents a comprehensive comparative analysis of various denoising techniques applied to CT, MRI and Ultrasound images. The evaluated techniques are categorized into three major groups:

spatial domain filters, edge-preserving filters and transform-based methods. The performance of these techniques is analyzed using standard image quality metrics: MSE, PSNR, SSIM and SNR.

1.1 Objectives

The specific objectives of this study are:

1. To analyze different categories of denoising techniques, including spatial domain filters, edge-preserving filters and transform-based methods.
2. To evaluate the performance of each filter under different noise conditions such as Gaussian noise, salt-and-pepper noise and speckle noise.
3. To quantitatively compare denoising performance using standard evaluation metrics including MSE, PSNR, SSIM and SNR.
4. To assess edge preservation and structural integrity after filtering, particularly in clinically significant regions.
5. To analyze computational complexity and processing time of each technique for practical medical applications.
6. To determine the suitability of each filtering method for specific medical imaging modalities such as MRI, CT and ultrasound.
7. To identify the most effective denoising strategy that balances noise suppression, structural preservation and computational efficiency.

2. Related Works / Literature Review

Medical image denoising has been widely studied due to its importance in improving diagnostic accuracy and preserving structural details. Various filtering, transform-based and learning-based techniques have been proposed for removing noise while maintaining important anatomical information.

Sagheer and George conducted a comprehensive review of medical image denoising algorithms used in different imaging modalities such as MRI, CT, PET and ultrasound. The study categorized denoising techniques into spatial filtering, transform-domain methods and statistical approaches. Their analysis highlighted that transform-based techniques such as wavelet transforms provide better noise suppression while preserving structural details in medical images.

Kaur and Dong presented a detailed survey on medical image denoising techniques and discussed different noise models affecting medical images, including Gaussian, Poisson, and speckle noise. The study emphasized the importance of selecting appropriate denoising methods based on the imaging modality and noise characteristics. Their work also highlighted the increasing role of machine learning and deep learning techniques in improving denoising performance.

Buades et al. introduced the Non-Local Means (NLM) algorithm for image denoising, which utilizes the redundancy present in natural images to remove noise while preserving fine structural details. The method computes the weighted average of pixels based on similarity between neighborhoods. Experimental results showed that NLM performs significantly better than classical filtering methods such as Gaussian smoothing and anisotropic diffusion in preserving edges and textures in MRI images.

Elhoseny et al. proposed an optimized bilateral filtering technique for medical image denoising combined with a convolutional neural network (CNN) classifier. The bilateral filter was optimized using bio-inspired algorithms to enhance noise removal while preserving edges. The results demonstrated improved PSNR and better classification accuracy compared to traditional filtering approaches.

Vilimek et al. performed a comparative analysis of wavelet transform-based filtering methods for ultrasound image denoising. The study showed that wavelet-based techniques effectively reduce speckle noise while preserving anatomical features and edges. The results confirmed that multi-resolution analysis provided by wavelet transforms improves denoising performance compared to conventional spatial filters.

Fan et al. presented a survey of classical and modern image denoising techniques, including spatial filtering, transform-based approaches and statistical models. The authors emphasized that no single denoising method performs best for all noise types and imaging modalities and that the selection depends on the characteristics of the noise and specific application requirements.

From the literature, it is evident that traditional spatial filters provide computational efficiency but may blur important edges, whereas transform-domain and non-local approaches achieve better structural preservation. Recent advances in deep learning further improve denoising performance but introduce higher computational complexity. Therefore, a systematic comparative analysis of classical and advanced filtering techniques remains important for identifying the most suitable denoising methods for different medical imaging modalities.

3. Methodology

The methodology adopted in this study involves two primary stages: Noise Modeling and the application of Filtering Techniques. These steps are designed to evaluate the effectiveness of various denoising approaches under controlled experimental conditions across three distinct medical imaging modalities.

3.1 Dataset

Experiments were conducted on three publicly available medical imaging datasets consisting of Computed Tomography (CT), Magnetic Resonance Imaging (MRI) and Ultrasound images. These datasets were selected to evaluate the proposed method across different medical imaging modalities and noise characteristics.

- MRI Dataset: Obtained from the Brain Tumor MRI Dataset available on Kaggle. It contains brain MRI scans of different tumor types along with non-tumor cases.
- CT Dataset: The COVID-19 CT Scan Lesion Segmentation Dataset, which includes CT scans with lesion annotations related to COVID-19 infection.
- Ultrasound Dataset: The Breast Ultrasound Images Dataset, which contains breast ultrasound images categorized into benign, malignant and normal classes.

3.2 Noise Modeling

To simulate realistic imaging conditions, three types of noise were artificially introduced into the clean images:

- Gaussian Noise (sigma = 0.01 to 0.1): A statistical noise model commonly observed in medical imaging systems due to electronic fluctuations during image acquisition. Noise values follow a normal (Gaussian) distribution, affecting almost every pixel with slight intensity variations. Widely used to simulate realistic degradation in MRI and CT images.
- Salt-and-Pepper Noise (density 0.01 to 0.05): Also referred to as impulse noise, it appears as randomly distributed bright and dark pixels caused by transmission errors,

sensor malfunction, or faulty memory locations. Some pixels are replaced with minimum (black) or maximum (white) intensity values.

- Speckle Noise: A multiplicative noise commonly found in ultrasound imaging, arising from the interference of reflected signals from multiple scatterers within biological tissues. It produces a granular appearance that reduces image contrast and obscures structural boundaries.

3.3 Filtering Techniques

Multiple filtering techniques were employed and categorized into three groups:

3.3.1 Spatial Domain Filters

- Mean Filter: A linear smoothing technique that replaces each pixel with the average intensity of its neighboring pixels within a predefined window. Effective at suppressing small fluctuations but may blur important edges.
- Moving Average Filter: Computes the average of pixel intensities over a sliding window. Simple and computationally efficient but may reduce image sharpness at edge regions.
- Exponential Moving Average (EMA): Assigns exponentially decreasing weights to older pixel values, retaining more information from nearby pixels compared to simple averaging.
- Gaussian Filter: Uses a Gaussian-shaped kernel for weighted averaging of neighboring pixels. Pixels closer to the kernel center receive higher weights, preserving general image structure better than mean filtering.
- Median Filter: A non-linear technique that replaces each pixel with the median value of its neighborhood. Highly effective for removing impulse (salt-and-pepper) noise while preserving edge information.
- Hybrid (Median + Mean): Applies median filtering first to remove impulse noise, followed by mean filtering to smooth remaining intensity variations. Combines advantages of both approaches.

3.3.2 Edge-Preserving Filters

- Bilateral Filter: Performs edge-preserving smoothing by combining spatial proximity and intensity similarity. Unlike traditional filters, it reduces noise while maintaining sharp edges.
- Anisotropic Diffusion: An iterative filtering process that selectively smooths homogeneous regions while preserving edge boundaries by restricting diffusion across strong intensity gradients.
- Non-Local Means (NLM) Filter: Exploits self-similarity within the image by comparing pixel neighborhoods across a larger search region and computing weighted averages based on patch similarity.

3.3.3 Frequency and Transform-Based Methods

- Wiener Filter: A statistical filtering technique that minimizes mean squared error between original and restored images by adapting to local image variance.
- Wavelet Transform: A multi-resolution technique that decomposes images into different frequency components. Noise reduction is achieved by thresholding high-frequency wavelet coefficients.
- Curvelet Transform: An advanced multi-scale transform designed to efficiently represent edges and curved structures by capturing directional information, well-suited for curved anatomical boundaries.

3.4 Evaluation Metrics

The following evaluation metrics were used to quantitatively assess the performance of the denoising methods:

- Mean Squared Error (MSE): Measures the average squared difference between the original and denoised image. Lower MSE values indicate better restoration quality.
- Peak Signal-to-Noise Ratio (PSNR): Derived from MSE, it represents the ratio between the maximum possible signal intensity and the noise level. Higher PSNR values indicate better image quality.
- Structural Similarity Index (SSIM): Evaluates image quality based on perceived changes in structural information, brightness and contrast. Values closer to 1.0 indicate higher structural similarity.
- Signal-to-Noise Ratio (SNR): Measures the strength of the desired signal relative to background noise. A higher SNR indicates that the image retains more useful information with less noise.

4. Tools and Technologies

The following tools and technologies were used in the implementation of this project:

4.1 Programming Language

- Python 3.x: The primary programming language used for all image processing, filtering and evaluation tasks, owing to its extensive scientific computing libraries.

4.2 Libraries and Frameworks

- NumPy: Used for efficient numerical computations and array operations on image data.
- OpenCV (cv2): Used for image loading, preprocessing, noise addition and filter application.
- SciPy: Provided implementations of Wiener filtering and statistical signal processing functions.
- scikit-image (skimage): Used for computing SSIM, PSNR and implementing Non-Local Means denoising.
- PyWavelets (pywt): Used for Wavelet Transform-based denoising operations.
- Matplotlib: Used for visualization of denoised images and graphical analysis of evaluation metrics.

4.3 Development Environment

- Jupyter Notebook / Google Colab: Used as the development and experimentation environment for running Python notebooks.
- Anaconda Distribution: Used for package management and environment setup.

4.4 Datasets

- Kaggle: Source for all three publicly available datasets used in experimentation (Brain Tumor MRI Dataset, COVID-19 CT Scan Dataset, Breast Ultrasound Images Dataset).

5. Results and Major Findings

The effectiveness of the applied denoising techniques was evaluated on three medical imaging modalities: CT, MRI and Ultrasound. These modalities exhibit different noise characteristics due to their distinct acquisition mechanisms. CT images typically contain Gaussian noise introduced during photon detection. MRI images often suffer from Rician or Gaussian noise. Ultrasound images are predominantly affected by speckle noise.

5.1 Spatial Domain Filters

The results across all three modalities demonstrate that spatial domain filters provide baseline denoising performance with varying degrees of effectiveness. For MRI images, the Median filter achieved the best quantitative performance, while the Hybrid filter showed the highest structural similarity. For CT images, the Moving Average filter performed optimally. For Ultrasound images, the Mean filter outperformed all others.

Table 1: Metrics for MRI Images - Spatial Domain Filters

Filter	MSE	PSNR (dB)	SNR (dB)	SSIM
Mean	108.9268	28.2329	16.6734	0.5974
Moving Average	150.9043	26.4059	14.8464	0.4486
Exponential Moving Average	280.8526	23.9107	12.3511	0.4110
Gaussian	92.7836	28.7824	17.2228	0.6023
Median	69.8764	30.0223	18.4628	0.7110
Hybrid (Mean + Median)	96.7328	29.1379	17.5783	0.7601

Table 2: Metrics for CT Images - Spatial Domain Filters

Filter	MSE	PSNR (dB)	SNR (dB)	SSIM
Mean	487.6832	21.3456	17.6724	0.3016
Moving Average	322.6276	23.1911	19.5178	0.3988
Exponential Moving Average	423.5661	21.9744	18.3011	0.4354
Gaussian	394.8692	22.2866	18.6133	0.3573
Median	358.3133	22.6979	19.0246	0.4035
Hybrid (Mean + Median)	474.5965	21.4512	17.7780	0.4099

Table 3: Metrics for Ultrasound Images - Spatial Domain Filters

Filter	MSE	PSNR (dB)	SNR (dB)	SSIM
Mean	87.6084	28.7610	21.0393	0.8124
Moving Average	122.7506	27.3399	19.6181	0.7674
Exponential Moving Average	146.5179	26.5091	18.7874	0.7308
Gaussian	394.8692	22.2866	18.6133	0.3573
Median	358.3133	22.6979	19.0246	0.4035
Hybrid (Mean + Median)	474.5965	21.4512	17.7780	0.4099

5.2 Edge-Preserving Filters

Among edge-preserving filters, the Bilateral filter consistently performed best across all modalities, achieving the lowest MSE and highest PSNR, SNR and SSIM values. Anisotropic diffusion performed poorly across all datasets with very high MSE (up to 2229.9798) and low SSIM (as low as 0.1191), indicating significant loss of structural information. Non-Local Means provided moderate performance.

Table 4: Metrics for MRI Images - Edge-Preserving Filters

Filter	MSE	PSNR (dB)	SNR (dB)	SSIM
Bilateral	101.8537	28.3259	16.7663	0.6007
Anisotropic Diffusion	1999.5718	15.1295	3.5699	0.1191
Non-Local Means	374.3960	22.4392	10.8796	0.3196

Table 5: Metrics for CT Images - Edge-Preserving Filters

Filter	MSE	PSNR (dB)	SNR (dB)	SSIM
Bilateral	234.1513	24.5306	20.8573	0.4204
Anisotropic Diffusion	1681.2194	15.8916	12.2183	0.2901
Non-Local Means	407.9758	22.0292	18.3559	0.4616

Table 6: Metrics for Ultrasound Images - Edge-Preserving Filters

Filter	MSE	PSNR (dB)	SNR (dB)	SSIM
Bilateral	94.1039	28.4429	20.7211	0.7730
Anisotropic Diffusion	2229.9798	14.6592	6.9374	0.1379
Non-Local Means	496.9549	21.30168	13.5799	0.5643

5.3 Frequency and Transform-Based Methods

The Wiener filter consistently outperformed other transform-based methods across all modalities, achieving the best balance of MSE, PSNR, SNR and SSIM. Wavelet transform provided moderate results with better structural similarity in some cases. Curvelet showed the weakest performance among transform-based methods.

Table 7: Metrics for MRI Images - Transform-Based Filters

Filter	MSE	PSNR (dB)	SNR (dB)	SSIM
Wiener	108.5705	27.8645	16.3049	0.5665
Wavelet	304.0011	23.3131	11.7535	0.3181
Curvelet	475.3059	21.3701	9.8105	0.2551

Table 8: Metrics for CT Images - Transform-Based Filters

Filter	MSE	PSNR (dB)	SNR (dB)	SSIM
Wiener	253.4505	24.2345	20.5612	0.3670
Wavelet	327.8870	22.9753	19.3020	0.4797
Curvelet	463.7230	21.4737	17.8004	0.4435

Table 9: Metrics for Ultrasound Images - Transform-Based Filters

Filter	MSE	PSNR (dB)	SNR (dB)	SSIM
Wiener	150.5482	26.3982	18.6765	0.7583
Wavelet	359.3280	22.7129	14.9912	0.5974
Curvelet	527.4858	21.0163	13.2945	0.4981

5.4 Summary of Key Findings

The comparative analysis across all three modalities and twelve filters leads to the following key findings:

- For MRI images, the Median filter achieves the best noise reduction performance (MSE = 69.8764, PSNR = 30.0223 dB, SNR = 18.4628), while the Hybrid (Mean + Median) filter provides the highest structural preservation (SSIM = 0.7601).
- For CT images, the Moving Average filter performs optimally in terms of reconstruction quality (MSE = 322.6276, PSNR = 23.1911 dB, SNR = 19.5178), while the Hybrid filter achieves the highest SSIM (0.4099).
- For Ultrasound images, the Mean filter outperforms all others across all metrics (MSE = 87.6084, PSNR = 28.7610 dB, SNR = 21.0393, SSIM = 0.8124).
- Among edge-preserving methods, the Bilateral filter consistently provides the best performance across all three modalities, achieving MRI PSNR = 28.3259 dB and SSIM = 0.6007 and Ultrasound SSIM = 0.7730.
- Among transform-based methods, the Wiener filter emerges as the most effective, with MRI PSNR = 27.8645 dB, CT PSNR = 24.2345 dB and Ultrasound PSNR = 26.3982 dB.
- Anisotropic diffusion exhibits the worst overall performance, with extremely high MSE values (up to 2229.9798 for Ultrasound) and very low SSIM (as low as 0.1191 for MRI), indicating severe structural degradation.
- No single filter performs optimally across all modalities, confirming that filter selection must be guided by the specific modality and noise characteristics.

6. Applications in the Medical Domain

Effective medical image denoising has significant practical implications across a broad range of clinical and computational applications:

- Tumor Segmentation in MRI: Cleaner MRI images facilitate more accurate delineation of tumor boundaries, improving pre-surgical planning and radiation therapy targeting.
- Skull Fracture Detection in CT: Improved CT image quality enhances the detection of hairline fractures and subtle skeletal abnormalities in trauma cases.
- Fetal Imaging Clarity in Ultrasound: Denoised ultrasound images improve the visibility of fetal anatomical structures, supporting more reliable prenatal diagnostics.
- Retinal Vessel Enhancement: Denoising applied to retinal imaging modalities facilitates clearer visualization of vascular structures for glaucoma and diabetic retinopathy screening.
- 3D Reconstruction and Implant Generation: High-quality denoised images are critical inputs for 3D volumetric reconstruction algorithms used in prosthetic design and surgical simulation.
- Computer-Aided Diagnosis (CAD) Systems: Improved image quality enhances the performance of downstream machine learning and deep learning-based diagnostic pipelines by reducing noise-induced misclassification.
- Radiological Reporting: Radiologists benefit from cleaner images that reduce interpretation ambiguity, leading to faster and more confident diagnoses.

Overall, effective denoising forms a foundational preprocessing step in modern medical image analysis workflows, enabling both manual diagnosis and automated AI-driven pipelines to operate more reliably.

7. Conclusion and Future Scope

7.1 Conclusion

This project presented a comprehensive comparative analysis of multiple denoising techniques applied to different medical imaging modalities, including CT, MRI and Ultrasound images. The evaluated techniques were categorized into three major groups: spatial domain filters, edge-preserving filters and frequency/transform-based methods. The performance of each method was quantitatively assessed using MSE, PSNR, SSIM and SNR.

The experimental results demonstrate that the performance of denoising algorithms strongly depends on the characteristics of the imaging modality and the type of noise present in the image. Spatial domain filters such as Mean, Gaussian and Median filters provide computationally efficient noise reduction and are suitable for basic smoothing tasks. However, these methods may lead to edge blurring and loss of important structural information in medical images.

Edge-preserving filters such as the Bilateral filter improve the preservation of structural features and anatomical boundaries while reducing noise. The Bilateral filter consistently outperformed all other edge-preserving methods across all modalities. Anisotropic diffusion showed the worst performance in this study and is not recommended for direct application without careful parameter tuning.

Transform-domain approaches, including the Wiener filter, Wavelet, and Curvelet transforms, demonstrated capabilities in capturing multi-scale image features and preserving structural details. The Wiener filter emerged as the most effective among transform-based methods across all modalities.

The key conclusion of this study is that no single denoising technique performs optimally for all imaging modalities. Instead, the selection of an appropriate denoising method should be guided by the noise characteristics and structural requirements of the specific medical imaging modality under consideration.

7.2 Future Scope

Although this study provides a comprehensive comparative analysis, several promising directions remain for future work:

- Adaptive Parameter Tuning: Developing mechanisms that automatically select filter parameters such as kernel size, diffusion coefficients and threshold values based on image-specific noise characteristics.
- Deep Learning-Based Denoising: Integrating and comparing the performance of deep learning architectures such as DnCNN, U-Net and ResNet-based denoising networks with traditional techniques to assess their clinical applicability.
- 3D Volumetric Denoising: Extending the current 2D framework to 3D volumetric denoising, which is critical for CT and MRI datasets that generate multi-slice volumetric data, preserving spatial continuity between slices.
- Real-World Clinical Validation: Testing these denoising methods on raw clinical datasets acquired from hospital imaging systems to validate performance under actual clinical conditions.
- Hybrid Deep-Classical Approaches: Exploring the combination of classical filtering techniques as preprocessing steps for deep learning pipelines to achieve improved denoising results with reduced computational overhead.

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Appendices

Appendix A: Abbreviations

Abbreviation	Full Form
MRI	Magnetic Resonance Imaging
CT	Computed Tomography
MSE	Mean Squared Error
PSNR	Peak Signal-to-Noise Ratio
SSIM	Structural Similarity Index
SNR	Signal-to-Noise Ratio
NLM	Non-Local Means
EMA	Exponential Moving Average
CAD	Computer-Aided Diagnosis
CNN	Convolutional Neural Network
IOHE	Industry Oriented Hands-On Experience

Appendix B: Noise Formulas

Gaussian Noise: Added as y = x + n, where n ~ N(0, sigma^2), x is the original image pixel value and y is the noisy pixel value.

Salt-and-Pepper Noise: Pixels are randomly set to 0 (pepper) or 255 (salt) with a given noise density probability p.

Speckle Noise: Modeled as multiplicative noise y = x + x * n, where n ~ N(0, sigma^2), reflecting the intensity-dependent nature of speckle.